Matthew Berger

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Senior Staff Rust engineer with 10+ years writing Rust, the last 5 years in production food assembly robotics. Founding software engineer at Hyphen, where I built the food assembly robotics controls suite from zero to production in 22 months, underpinning $35M+ raised from Chipotle and Cava. Background spans embedded firmware to cloud, with prior production experience in aerospace imaging (Sierra Nevada) and safety-critical medical robotics (Hamilton). Author of Rust crates (freecs, enum2 family) with 280k+ combined crates.io downloads.

Work Experience

Hyphen (Culinary Robotics) — Remote
Senior Staff Software Engineer (Founding Engineer) — *January 2026 – May 2026*

- Shipped Hyphen Studio, a Dioxus desktop application enabling operators to configure makeline layouts and parameters directly
- Replaced TypeScript/GraphQL tablet portal with a pure Rust Leptos rewrite connecting directly to the IPC broker, eliminating an entire backend tier

Hyphen (Culinary Robotics) — Remote
Staff Software Engineer (Founding Engineer) — *September 2022 – January 2026*

- Sole architect of the production food assembly robotics controls suite, merging 1,000+ pull requests over 4 years across firmware, controls, IPC, motion, and cloud infrastructure
- Designed and built the production async pub/sub message broker over TCP with cross-host bridging, coordinating every distributed process across the industrial PC and embedded boards, at sub-millisecond local latency and zero message loss across 3+ years of continuous fleet operation
- Built multi-axis motion control with trajectory planning, kinematics, and motor drivers with closed-loop PID for MG4010 and LKMG actuators
- Built the full stack from bare-metal firmware (RP2040/Embassy) to distributed robotics control system, custom Yocto Linux distribution, and AWS cloud
- Built Hyphen Explorer (Bevy + egui) for real-time IPC visualization, debugging, and system control, achieving 100% adoption across all Hyphen engineers
- Mentored team of web developers into Rust systems programmers through architectural review, pair programming, and documented patterns

Hyphen (Culinary Robotics) — Remote
Senior Software Engineer (Founding Engineer) — *July 2021 – September 2022*

- Diagnosed scaling limitations in legacy PLC system preventing growth beyond prototype
- Led technical pivot from costly PLC-based control system to embedded Rust architecture
- Pioneered RP2040 firmware with Embassy-rs, creating first Rust TMC5160 motor and AS5048A encoder drivers proving Rust could meet real-time control loop requirements

Sierra Nevada Corporation — Englewood, CO
Software Engineer III — *May 2020 – July 2021*

- Developed C++/Rust aerospace imaging system (5GB/sec) and simulator saving 3 months on delivery timeline

Scientific Games — Reno, NV
Software Engineer — *July 2019 – May 2020*

- Shipped performance and gameplay defect fixes in a Unity-based casino platform

Hamilton Company — Reno, NV
Software Engineer (Intern: October 2014 – December 2017) — *January 2018 – July 2019*

- Built safety-critical software for liquid-handling medical robots

- Automated gravimetric analysis QA process, saving 1+ week per robot across production fleet
- Reduced calibration development from 2 months to 2 weeks through reusable plugin framework
- Developed calibration application driving Hamilton's firmware via UI for Illumina, Hamilton's largest OEM customer ($2M in machine purchases)

Skills

Languages: Rust, C++, C, C#, TypeScript, Python
Systems: Real-time low-latency systems, embedded (RP2040, Embassy), async Rust (Tokio, async-std), IPC, message brokers, networking, Yocto Linux, distributed systems
Cross-Platform: Linux, macOS, Windows, Android, WebAssembly, Steam Deck, OpenXR/VR
UI/Graphics: Leptos, Dioxus, egui, Bevy, wgpu, Vulkan, DX12, Metal, OpenGL
Cloud/Infra: AWS, Pulumi, Greengrass IoT, Docker, GitHub Actions

Open Source

- **Nightshade**: data-oriented Rust game engine featuring a custom render graph, built-in editor, rapier3d physics, kira audio, gltf asset loading, procedural terrain, and Steam integration. Runs natively on DX12/Metal/Vulkan and in-browser via WebGPU (live demo: matthewberger.dev/nightshade)
- **freecs** (48k+ downloads on crates.io): archetype-based ECS in Rust with zero unsafe code, Rayon-parallel system execution, sparse-set tags, deferred command buffers, and watermark change detection. The entire ECS is generated at compile time via a declarative macro. Design documented in a 3-part series at matthewberger.dev/articles
- **enum2 macro family** (enum2str, enum2contract, enum2egui, enum2pos, enum2repr): derive macros for enum-driven design patterns with 230k+ total downloads on crates.io
- **wgpu-example**: minimal Rust + wgpu + egui template showing how to build graphics that run everywhere, including native desktop, WebGPU and WebGL via WebAssembly, Android, Steam Deck, and OpenXR VR with hand tracking
- **cameras** (new in 2026): cross-platform Rust camera library with data-oriented design (explicit format negotiation, push-based frame delivery, zero trait objects) designed for real-time video frame pipelines across Linux, macOS, and Windows. Ships with Dioxus and egui integrations

Education

University of Nevada, Reno 2017
B.S. Computer Science & Engineering, Minor in Mathematics